INFO 4220 – Networks II: Market Design
Spring 2024 – Problem Set 1

Due date: (Wednesday) February 7, 11.59PM EST

Late submissions will be penalized with 1 point for every hour late.

Instructions: For each question, show all your work or you will lose some or all of the credit for that question. ***Just showing the result without showing and explaining all steps you took to find it does not count as fully correct answer.***

How to submit: Solutions must be submitted using Gradescope. For instructions on how to submit the assignment, go to this guide or watch this video.

Important: You must select the pages for each question in Gradescope. Not doing this will lead to delays and errors in your grades. If you don't know how to do this, read this guide.

Asking questions in Ed Discussions: You can use Ed Discussions to ask clarification questions about the problems. You cannot ask on Ed Discussions how to solve a particular problem, and especially, you cannot post on Ed Discussions (or anywhere else) your solution to the problem set (or a subset of it). Disclosing the solution of problems in Ed Discussions (or anywhere else) will be considered an academic integrity violation and reported accordingly. If you don't know how to solve a problem, don't ask on Ed Discussions, come to office hours! (Although, we will not solve the problem for you)

Who did you work with to solve this PS, if anyone (remember each of you must turn in their own solution, which has to be your own original work)? (0 pt)

Are you using and slip day for this PS? Remember that you have 2 for the entire semester, and they can only be used in 1-day increments. (0 pt)

Problem 1 (6 points, 2 pts each)

Alex and Jamie started dating in November 2023 and they are now deciding how to spend their first Valentine's Day together. They can choose between Soccer Game, Cinema, Dinner, or Staying at Home. Alex would be very excited about going to a Soccer game, while Jamie would be very happy to go to the cinema and watch a romantic movie. Even if their preferences differ, they need to make a decision and coordinate their plans. To avoid conflicts and a tiring negotiation process, they have agreed they will both privately write their decision on a piece of paper and reveal it at the same time. The payoffs are reported in the matrix below (which is known to both Alex and Jamie).

		Jamie			
		Soccer	**Cinema**	**Dinner**	**Home**
Alex	**Soccer**	(100, -20)	(-30, -30)	(20, 0)	(0, -10)
	Cinema	(-30, -30)	(-20, 100)	(0,20)	(-10, 0)
	Dinner	(20, 0)	(0, 20)	(50, 50)	(0, 0)
	Home	(0, -10)	(-10, 0)	(0, 0)	(10, 10)

a) Iteratively delete all dominated strategies and show the resulting payoff matrix.

We can iteratively eliminate **strictly dominated strategies**. A strategy is strictly dominated for player i if there exist another strategy that gives them an stricter higher payoff for any strategy of player $-i$.

Step 1 – Cinema is strictly dominated for Alex. In fact, for Alex, Dinner always provides a strictly higher payoff. We cancel it.

		Jamie			
		Soccer	**Cinema**	**Dinner**	**Home**
Alex	**Soccer**	(100, -20)	(-30, -30)	(20, 0)	(0, -10)
	Cinema	(-30, -30)	(-20, 100)	(0,20)	(-10, 0)
	Dinner	(20, 0)	(0, 20)	(50, 50)	(0, 0)
	Home	(0, -10)	(-10, 0)	(0, 0)	(10, 10)

Now that we have eliminated Cinema for Alex, we have that:

Step 2 – Soccer is strictly dominated for Jamie. In fact, Dinner always gives Jamie a strictly higher payoff that Soccer. We eliminate Soccer for Jamie.

		Jamie			
		Soccer	**Cinema**	**Dinner**	**Home**
Alex	**Soccer**	(100, -20)	(-30, -30)	(20, 0)	(0, -10)
	Cinema	(-30, -30)	(-20, 100)	(0,20)	(-10, 0)
	Dinner	(20, 0)	(0, 20)	(50, 50)	(0, 0)
	Home	(0, -10)	(-10, 0)	(0, 0)	(10, 10)

We are left with the 3x3 matrix.

		Jamie		
		Cinema	**Dinner**	**Home**
Alex	**Soccer**	(-30, -30)	(20, 0)	(0, -10)
	Dinner	(0, 20)	(50, 50)	(0, 0)
	Home	(-10, 0)	(0, 0)	(10, 10)

Note 1: this is the correct solution of iterative elimination of dominated strategies which requires *for a given player in the game*, that *an action can be iteratively eliminated at each step if and only if there exists another action that provides that fixed player with a strictly higher payoff for any other action of the coplayer. Formally a strategy* $\tilde{s}^i \in S^i$ *is a dominated strategy for player* i *if there exist another strategy* s^i *such that:*

$$\pi^i\left(\tilde{s}^i, s^{-i}\right) < \pi^i\left(s^i, s^{-i}\right), for\ every\ s^{-i} \in S^{-i}$$

In this example, 'Cinema' is only **weakly dominated by 'Dinner' --> weakly because, when Alex chooses 'Home', Cinema and Dinner provide both a payoff equal to zero to Jamie.**

Note 2: The elimination of dominated strategies is defined as deleting strictly dominated strategies. However, we are assigning full credit to those students who have eliminated Soccer for Alex and Cinema for Jamie. The logic for eliminating those strategies is that while they are only weakly dominated, they are never best response to an action of Jamie, and thus will never be part of a Nash Equilibrium (it is never rational to play them).

		Jamie	
		Dinner	**Home**
Alex	**Dinner**	(50, 50)	(0, 0)
	Home	(0, 0)	(10, 10)

However, **this is not strictly correct** if we are iteratively eliminating dominated strategies. Just wanted to clarify this, as I did not explain this clearly when I explained the iterative elimination of dominated strategies.

b) Is there any Nash equilibrium in pure strategies? If you find one (or more), use the definition of Nash Equilibrium to prove it is an equilibrium.

There are two pure strategy equilibria in this game: (Dinner, Dinner) and (Home, Home). In both these scenarios, neither player would benefit from changing their strategy while the other's remains the same. In terms of the definition.

		Jamie		
		Cinema	**Dinner**	**Home**
Alex	**Soccer**	(-30, -30)	(20, 0)	(0, -10)
	Dinner	(0, 20)	(**50**, **50**)	(0, 0)
	Home	(-10, 0)	(0, 0)	(**10**, **10**)

To prove that (Dinner, Dinner) is a N.E. we need to show that playing 'Dinner' gives Alex (or Jamie) a higher payoff that any other faction fixing the action of Jamie (or Alex). Namely, we need that:

For Alex, assuming that Jamie will play 'Dinner'
$\pi_{Alex}(Dinner,\ Dinner) \geq \pi_{Alex}(Home,\ Dinner)$
$\pi_{Alex}(Dinner,\ Dinner) \geq \pi_{Alex}(Soccer,\ Dinner)$
$\pi_{Jamie}(Dinner,\ Dinner) \geq \pi_{Jamie}(Dinner,\ Home)$
$\pi_{Jamie}(Dinner,\ Dinner) \geq \pi_{Jamie}(Dinner,\ Cinema)$

To prove that (Home, Home) is a N.E. we need to show these conditions are true:
$\pi_{Alex}(Home, Home) \geq \pi_{Alex}(Dinner, Home)$
$\pi_{Alex}(Home,\ Home) \geq \pi_{Alex}(Soccer,\ Home)$
$\pi_{Jamie}(Home, Home) \geq \pi_{Jamie}(Home, Dinner)$
$\pi_{Jamie}(Home,\ Home) \geq \pi_{Jamie}(Home,\ Cinema)$

Alternatively, it can be written in terms of best responses:
$BR_{Alex}(Dinner)\ =\ BR_{Jamie}(Dinner) = Dinner$
$BR_{Alex}(Home)\ =\ BR_{Jamie}(Home) = Home$

0.5 points for each equilibrium. 1 point for proving it is an equilibrium.

c) Is there any strategy profile that is pareto optimal in this game? Do you expect that to be the outcome of this game? Discuss.

In the Valentine's Day game, the two equilibria identified were both choosing Dinner and both choosing Home. To determine if one of these is Pareto optimal, we compare them to see if one can be improved without making the other worse off:

- Dinner (50, 50): Both Alex and Jamie have a good level of satisfaction.
- Home (10, 10): Both have a lower level of satisfaction, but it's a safe choice.

Comparing these, the (Dinner, Dinner) equilibrium is Pareto optimal because moving from (Dinner, Dinner) to (Home, Home) (or to any other outcome) would make both Alex and Jamie worse off. In the (Dinner, Dinner) outcome, there's no way to make one person happier without making the other unhappier, which fits the definition of Pareto optimality. The (Home, Home) scenario, while a Nash Equilibrium, is not Pareto optimal because there is an alternative (Dinner, Dinner) that would make both players better off. While game theory does not make any predictions regarding which equilibrium will be chosen, it is reasonable to expect that both players will realize (Dinner, Dinner) is a best a

1.5 points for explaining why (Dinner, Dinner) is Pareto optimal and 0.5 points for explaining that it is the likely outcome of the game.

Problem 2 (6 points, 3 pts each)

There are two firms, A and B, operating in the same market (they both produce and sell a homogeneous product). The two firms need to choose the prices they will charge for their product. When making this decision, the two firms know that if they choose a price higher than the one of their competitor, they

will not make any profit as all consumers will go for the cheaper option of the competitor (which is the cheapest option in the market for that good as these are the only two firms selling it)[1].

a) Suppose that the firms choose their prices simultaneously. They will learn the price of their competitor only after they have made a choice themselves. The firms can choose between a high price H, a medium price M, or a low price L. The matrix below shows the payoffs of this simultaneous game. Find the Nash Equilibrium in this game.

		Firm B		
		L	M	H
Firm A	L	(20, 20)	(30, 0)	(40, 0)
	M	(0, 30)	(25, 25)	(35, 0)
	H	(0, 40)	(0, 35)	(30, 30)

L is a dominant strategy for both firm A and firm B as choosing a low price (L) ensures them to earn higher profits regardless the strategy of their competitor. For instance, if firm B chooses (M), firm A will choose (L) as it will earn 30 (>25, i.e., profits firm A gets if chooses (M) when firm B chooses (M); and >0, the profits firm A will get when choosing (H) when firm B chooses (M)). Therefore, the unique Nash Equilibrium of this game is represented by the strategy profile (L,L), where both firms choose to set low prices. In fact, if any of the two firms chooses a higher price, the other firm always has the unilateral incentive of charging a lower price and capturing the whole market.
2 points for identifying the equilibrium (L, L) and 1 point for explanation.

b) Suppose that the CEO of firm A tries to give the CEO of firm B a call to decide their prices together (although in many places this would be against the law). After they agree on a price during the call, they still need to set their price in the market, and they will only learn the price set by the other firm after they reveal their own. If this game happens only once, what does game theory predict will happen? Explain your answer.

They both agree in the call to play (H, H). 1 point.
After the call, since it is a game that happens only once, both will have incentives to deviate, leading to the equilibrium of (L, L). 2 points.

Problem 3 (6 points, 3 points each part)

Alice is an investor deciding whether to fund or not Bob's project. Alice doesn't know if Bob will work hard or not if he gets the investment. A random draw will determine the signal that Bob gets. With probability 1/2 Bob will get the signal "hard worker" (this probability is known to Alice). Otherwise, he will get the signal "not a hard worker". The payoffs each receive depends on this signal and are shown in the payoff matrix below:

Bob is a hard worker			
		Bob	
		Work	Relax
Alice	Invest	9, 10	3, 4
	Pass	5, 3	5, 4

Bob is not a hard worker			
		Bob	
		Work	Relax
Alice	Invest	6,5	1, 4
	Pass	5, 0	5, 3

[1] In the economic literature, it is often referred to this kind of problems as *competition a la Bertrand*

a) Explain what constitutes a Bayesian Nash Equilibrium (BNE) in this game. (Hint: Think what each of the players know, and what they have to decide based on this information)

Each player needs to find a strategy that is the best response to the other player's strategy. For Bob a strategy consists on (1) what he will do if he is a hard worker and (2) what he would do if he is not a hard worker. 2 **points**

For Alice, a strategy is constituted by her choice of one action. Therefore, we can write a strategy profile in this way:
$(Alice's\ action;\ (Bob's\ action\ if\ he\ is\ HW;\ Bob's\ action\ if\ he's\ NHW))$
1 point.

b) Find all possible BNE in this game.

There are eight possible strategy profiles:
$(Invest;\ (W,W))$
$(Invest;\ (W,R))$
$(Invest;\ (R,W))$
$(Invest;\ (R,R))$
$(Pass;\ (W,W))$
$(Pass;\ (W,R))$
$(Pass;\ (R,W))$
$(Pass;\ (R,R))$

0.5 points
What is the utility associated to each of the strategy profiles? Let's describe them as u(strategy profile) = (Expected utility of Alice; utility of Bob). Notice that **the expected utility for Alice,** for a fixed Alice's action and a strategy profile is the average (because the probability of Bob being a hard worker is 0.5) of the payoff that she would obtain in the two scenarios (hard worker, not hard worker). We have that:
$(u(Invest;\ (W,W)) = (9 \cdot 0.5 + 6 \cdot 0.5 = 7.5;\ 10; 5)$
$u(Invest;\ (W,R)) = (9 \cdot 0.5 + 1 \cdot 0.5 = 5;\ 10; 4)$
$u(Invest;\ (R,W)) = (3 \cdot 0.5 + 6 \cdot 0.5 = 4.5;\ 4; 5)$
$u(Invest;\ (R,R)) = (3 \cdot 0.5 + 1 \cdot 0.5 = 2;\ 4; 4)$
$u(Pass;\ (W,W)) = (5 \cdot 0.5 + 5 \cdot 0.5 = 5;\ 3; 0)$
$u(Pass;\ (W,R)) = (5 \cdot 0.5 + 5 \cdot 0.5 = 5;\ 3; 3)$
$u(Pass;\ (R,W)) = (5 \cdot 0.5 + 5 \cdot 0.5 = 5;\ 4; 0)$
$u(Pass;\ (R,R)) = (5 \cdot 0.5 + 5 \cdot 0.5 = 5;\ 4; 3)$
0.5 points

The best response of Bob when Alice plays 'Invest', is the strategy profile (W,W). In fact, this strategy profile gives Bob the highest payoff among the other possible strategy profiles. He gets 10 if he is hard worker, 5 if he's not. Similarly, we can define best responses of Bob for other actions played by Alice, and best responses of Alice for any strategy profile played by Bob. We have that:

$$BR_{Bob}(Invest) = (W,W)$$
$$BR_{Bob}(Pass) = (R,R)$$

0.5 points

We now know that (W,W) and (R,R) are the only best responses for Bob. So we consider the best response of Alice to those only – we don't need to see what's Alice's best response to other strategy profiles as they cannot constitute part of the Bayes Nash Equilibrium by definition.

$$BR_{Alice}(W,W) = Invest$$
$$BR_{Alice}(R,R) = Pass$$

0.5 points

There are two possible BNEs:

$$(Invest;\ (W,W)) \text{ and } (Pass;\ (R,R))$$

1 point

Problem 4 (5 points, 1 point each)

The release of the 2023 Barbie movie brings about significant changes in the Barbie merchandise market. Retailers and producers are adjusting their strategies in response to the movie's impact on consumer demand. The following table presents the relationship between the price of Barbie merchandise and the quantity supplied and demanded:

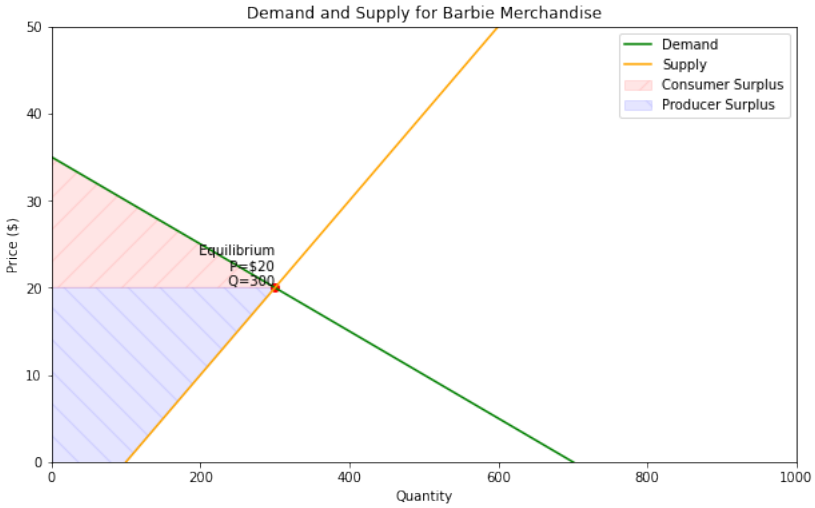
Price (in $)	Demand (in units)	Supply (in units)
10	500	200
15	400	250
20	300	300
25	200	350
30	100	400

a) Find the linear equations for the demand and supply curves for Barbie merchandise that fit the values in the table.

Demand: $Q_d = a - bP$.
Supply: $Q_s = c + dP$.
Using table values, $Q_d = 700 - 20P$ and $Q_s = 100 + 10P$.
0.5 points each equation.

b) On a graph, illustrate the demand and supply curves. Find the market equilibrium and calculate the consumer and producer surplus. Show the demand and supply curves, and the consumer and producer surplus in a graph.

Equilibrium: Solve $Q_d = Q_s$.
Equilibrium: P = $20, Q = 300 units . (**0.25 points**)
Consumer surplus = (35-20) * 300/2= 2250 (**0.25 points**)
Producer surplus = (100+300) *20/2= 4000 (**0.25 points)**



0.25 points.

c) With the release of the Barbie movie, demand increases by 50% at each price point. How does the equilibrium price compare to that in part (b)?

New demand curve: $Q_d' = 1.5 * (700 - 20P) = 1050 – 30P$ (**0.3 points**).
New equilibrium: P = $23.75, Q = 337.5 units (**0.3 points**).
Price increases (**0.4 points**).

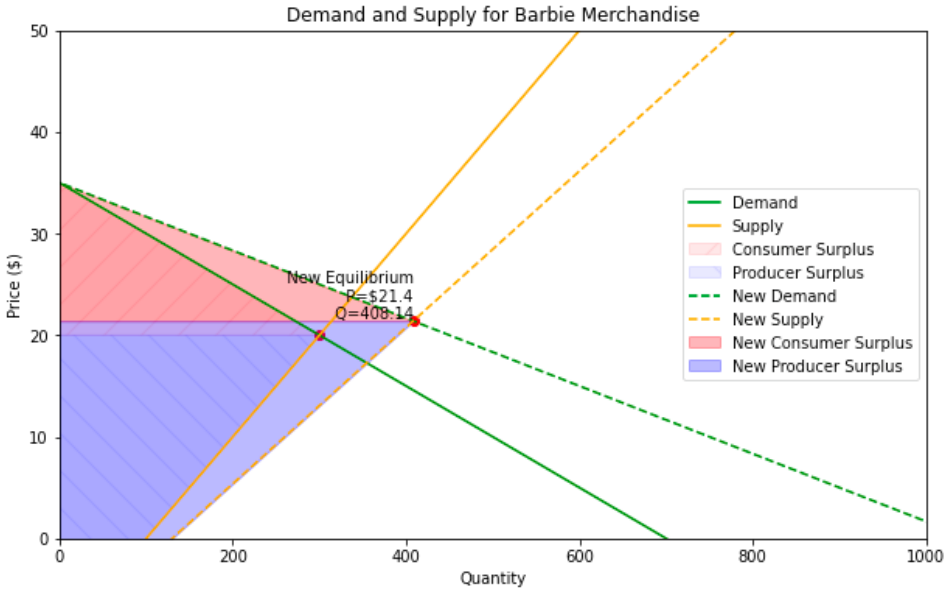
d) Assuming the movie's success and increased leads to an increase in production efficiency and at each price point in the table 30% additional units of Barbie merchandise are offered. Find the new supply curve and the new market equilibrium (use the demand curve from part (c)).

New supply curve with increased efficiency: $Q_s' = 1.3 * (100 + 10P) = 130 + 13P$ (**0.5 points**).
New equilibrium: P = $21.4, Q = 408.14 units (**0.5 points**).

e) Assess the impact of the movie on economic welfare: Compare changes in consumer and producer surplus before (i.e. equilibrium from part b) and after the movie release (i.e. equilibrium for part d). Who benefits the most from this market change?

Calculate and compare new CS and PS with initial values. Determine the primary beneficiary of the market change (consumers or producers).
Initial CS: $2250
Initial PS: $4000

New demand curve (from c): $Q_d' = 1.5 * (700 - 20P) = 1050 – 30P$
New supply curve (from d): $Q_d' = Q_s = 1.3 * 100 + 13P = 130 + 13P$
New equilibrium: P = $21.4, Q = 408.14 units
New CS: (35-21.4) * 408.14/2= 2775.4 **(0.25 points)**
New PS: (408.18+130) * 21.4/2= 5758.1 (**0.25 points**)



With the market change, consumer surplus increases and producer surplus increases. Producers benefits more from the market change as more benefits were captured by the huge increase from the demand side. (**0.5 points**)

Problem 5 (7 points)

In a salsa dancing class, eight students need to pair up to dance salsa. Four students (A_1, A_2, A_3, A_4) will lead, and the other four (B_1, B_2, B_3, B_4) will follow. Students' preferences over whom they prefer to dance with are as follows:

Students Leading				**Students Following**			
A_1	A_2	A_3	A_4	B_1	B_2	B_3	B_4
B_1	B_4	B_1	B_2	A_2	A_2	A_3	A_3
B_4	B_2	B_3	B_4	A_3	A_4	A_2	A_4
B_3	B_1	B_4	B_3	A_4	A_1	A_4	A_1
B_2	B_3	B_2	B_1	A_1	A_3	A_1	A_2

a) Use the deferred acceptance algorithm with "students following" proposing to find dancing partners. (2 pts)

1st round
B1 →A2
B2 → A2
B3 → A3
B4 → A3
A1 accepts B2 and rejects B1.
A3 accepts B3 and rejects B4.
0.5 points

2nd round
B1 →A3
B4 → A4
A2 accepts B1 and rejects B3.
A4 accepts B4
0.5 points

3rd round
B3 → A2
A2 rejects B3

4th round
B3 → A4
A4 rejects B3

5th round
B3 →A1
A1 accepts B3.
0.5 points

Matching: A1 - B3, A2 - B2, A3 - B1, A4 - B4
0.5 points

b) Is there any stable matching that the "students leading" would prefer? (2 pts)

The easiest way of proving this is doing D.A. with leading students proposing and see if the result is different.

1st Round
A1 → B1
A2 → B4
A3 → B1
A4 → B2
B1 rejects A1.
0.5 points.

2nd Round
A1 →B4
B4 rejects A2.

3rd Round
A2 →B2
B2 rejects A4.
0.5 points.

4th Round
A4 →B4
B4 rejects A1.

5th Round
A1 → B3

Matching is A1 - B3, A2 - B2, A3 - B1, and A4 - B4 which is the same as in a), then it is the only stable matching.
1 points.

c) The two teachers decide to join the students to show some new moves. One of them will take a leading role (A_5) y the other a follower (B_5). They don't want to have to do the matching process again, so they decide they will start from the current matching (from part a) and B_5 will ask leading students if they want to change their current partner for them. B_5 will propose first to their most preferred "student leding". If rejected, they next propose to their most preferred students among those they have not proposed yet. When a "student leading" accepts an offer, the "student following" that was matched to them can make a new offer. The following happens:
   - B_5 proposes to A_1 and is rejected.
   - B_5 proposes to A_2 and is accepted. Now B_2 no longer has a partner and can make a new proposal.
   - B_2 proposes to A_4 and is accepted. Now B_4 no longer has a partner.
   - B_4 proposes to A_1 and is accepted. Now B_3 no longer has a partner.
   - B_3 proposes to A_5 and is accepted. The process ends.

   What can we learn about students' preferences for teachers as dance partners from observing this process? And what can we learn about teachers' preferences for students as dance partners? Will the matching that results from this process be stable? Why or why not? (Hint: Think on the definition of stability and what you learnt about students and teachers' preferences) (3 pts)

B5 → A1 and is rejected: A1 likes b3 better than B5, we do not know about B2.
0.5 points.
B5 → A2 and is accepted: A2 likes B5 better than B2, B1, B3. We don't know who wins between B4 and B5.
0.5 points.
B2 →A4 accepted: We do not learn anything nee other than B2 likes A4 better than A5.
0.5 points.
B4 → A1 is accepted. B4 likes A1 better than A5.
B3 → A5: Nothing new.
The final matching is: B1 – A3, B2 – A4, B3 – A5, B4 – A1, B5 – A2
0.5 points.

It is not stable because there can be a blocking pair. We don't know anything about the preferences of A5, and we don't know where A5 stands in the preferences of B1. Thus, it could be the case that B1 is the most preferred for A5, and A5 is the most preferred for B1. If this is the case A5 – B1 would form a blocking pair.
1 point.